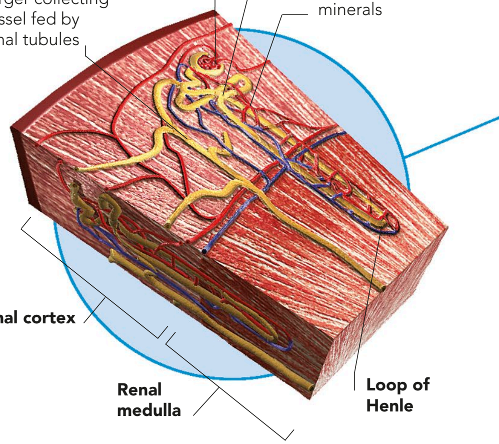
KIDNEY STRUCTURE

THE KIDNEYS ARE A PAIR OF ORGANS SITUATED EITHER SIDE OF THE SPINAL COLUMN AND AT THE UPPER REAR OF THE ABDOMINAL CAVITY. THEY FILTER WASTE PRODUCTS FROM THE BLOOD AND EXCRETE THEM, ALONG WITH EXCESS WATER, AS URINE.

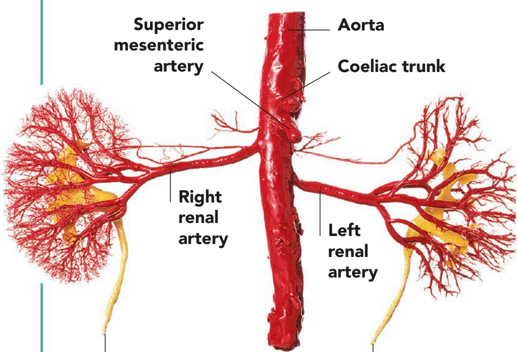
INSIDE THE KIDNEY

Each kidney is protected by three outer layers: a tough external coat of fibrous connective tissue, the renal fascia; a layer of fatty tissue, the adipose capsule; and inside this, another fibrous layer, the renal capsule. The main body of the kidney also has three layers: the renal cortex, which is packed full of knots of capillaries known as glomeruli and their capsules; next, the renal medulla, which contains capillaries and urine-forming tubules; and a central space where the urine collects, known as the renal pelvis. The glomeruli, capsules, and tubules are the constituent parts of the kidney's million-plus microfiltering units, called nephrons.



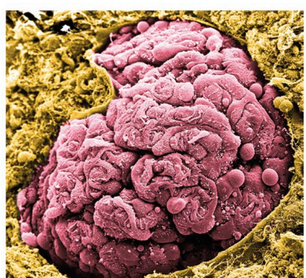
NEPHRON

Each microfiltering unit, or nephron, spans the cortex and medulla. The glomerulus, capsule, proximal and distal tubules, and the smaller urine-collecting ducts are in the cortex. The medulla contains mainly the long tubule loops of Henle and the larger urine-collecting ducts.



BLOOD SUPPLY TO THE KIDNEYS

The left and right renal arteries are branches of the aorta, which carries blood directly from the heart. The arteries leave the coeliac trunk of the aorta just below the superior mesenteric artery. The renal arteries form a branching network that supplies blood to the kidneys.



GLOMERULUS

This microscope image shows the tangled system of a glomerulus (pink). A filtrate fluid oozes from the glomerulus and is collected by the cup-like Bowman's capsule (brown).